

SENTINEL

Sentinel audits a Sparkplug estate's broker, certificates, identities and topic permissions and grades what it finds, read-only, with evidence and a fix for every finding.

— VERSION 0.1.0 IGNITION 8.1.19+ AND 8.3 RUNS ON: ANY HOST WITH BROKER ACCESS

CONTENTS

1. [1. Overview](#)
2. [2. What You Need](#)
3. [3. Install](#)
4. [4. License](#)
5. [5. Configure](#)
6. [6. Operate](#)
7. [7. Verify](#)
8. [8. Troubleshoot](#)
9. [9. Reference](#)
10. [10. Support](#)

1. OVERVIEW

Sentinel audits a Sparkplug estate's posture and grades it without ever touching it. It connects to the broker, listens for a bounded window, optionally runs active probes, and produces a scored report where every finding carries the frames or config lines that prove it and a remediation that names the setting. It exposes no publish method at all; it cannot write to the estate.

Sentinel is vendor-neutral: it audits any Sparkplug estate, not only one running SparkBridge.

- A SparkBridge Host gateway for the estate being audited.
- The signed Sentinel release from the SparkBridge bundle (built as `sparkaudit-<version>-all.jar`).
- Java 17 if building from source.
- Broker credentials for the auditor identity; optionally the broker's `credentials.xml` and a posture properties file for config-derived findings.

3. INSTALL

Sentinel carries its own license. Install the signed release through Config > Modules > Install or Upgrade a Module; no build step is required for the packaged release.

To build from source: requires Java 17 and `io.sparkbridge:common` published to `mavenLocal`.

```
JAVA_HOME=~/.local/opt/jdk-17.0.19+10/Contents/Home ./gradlew fatJar
java -jar build/libs/sparkaudit-0.1.0-all.jar --help
```

Soft-licensed; Sentinel runs in full whether or not a key is present. License file lookup order: `--license <file>`, the `SPARKBRIDGE_LICENSE` environment variable, then `sparkbridge-license.key` in the working directory. Run `sparkaudit license` for full status.

5. CONFIGURE

Sentinel runs as a passive audit by default:

```
java -jar sparkaudit-0.1.0-all.jar \
  --broker tcp://broker.example:1883 --username auditor \
  --password-file secrets.properties audit.password \
  --estate "Plant 4" --window 300 --expect-groups Plant4,Utilities \
  --out-html report.html --out-json report.json
```

<code>--CREDENTIALS-XML, - -POSTURE</code>	Add config-derived findings the wire alone cannot show.
<code>--ALLOW-ACTIVE- PROBES</code>	Enables active probes. Off by default.
<code>--TLS-LISTENER</code>	Checks a TLS listener.
<code>--PROFILE FREE / -- PROFILE FULL</code>	<code>free</code> withholds full evidence for a pre-sales report; <code>full</code> (default) includes complete evidence and remediation.
<code>--CAPTURE-OUT CAPTURE.JSONL</code>	Writes the raw window to disk.
<code>--REPLAY CAPTURE.JSONL</code>	Re-evaluates rules offline against a captured window, no broker involved.

The four axes Sentinel scores: connection posture, and (per the module's own documentation) further axes covering broker configuration, identity and permission hygiene, and topic/namespace isolation. Active probes only run with `--allow-active-probes`; with probes off, the connection-posture axis reports that it was not exercised.

Sentinel is a point-in-time audit tool, not a running service. Run it on demand or on a schedule against an estate's broker. It never publishes to any topic and exposes no publish method. Use `--capture-out` to keep a raw record of any audit window so a finding stays checkable later, or by someone who was not in the room, or so a customer can send an estate SparkBridge is not allowed to connect to.

7. VERIFY

Run the audit and check the exit code: 0 clean, 1 findings below critical, 2 at least one critical finding, 3 the audit could not run. Use `--capture-out capture.jsonl` and later `--replay capture.jsonl` to confirm the same findings reproduce offline against the same window.

SYMPTOM	CAUSE	FIX
	The audit could not run (bad broker URI, bad credentials, or the window elapsed with no connection)	Confirm <code>--broker</code> , <code>--username</code> and the password file path are correct
	Active probes are off by default	Add <code>--allow-active-probes</code> if authorized to probe the estate
	<code>--credentials-xml</code> or <code>--posture</code> not supplied	Pass the broker's real <code>credentials.xml</code> and a posture properties file
	<code>io.sparkbridge:common</code> not published to mavenLocal	Run <code>./gradlew :common:publishToMavenLocal</code> in the SparkBridge repo first

9. REFERENCE

- Exit codes: 0 clean, 1 sub-critical findings, 2 critical finding, 3 audit could not run.
- License file: `sparkbridge-license.key` (working directory, `--license`, or `SPARKBRIDGE_LICENSE`).
- Build artifact: `sparkaudit-0.1.0-all.jar`.